TAP transport cycle in six structures

TAP (transporter associated with antigen processing) moves peptides into the endoplasmic reticulum (ER). Read clockwise from 1.

NBD: nucleotide-binding domain. Dashed line: NBD gap, Cα distance from TAP1 Asp674 (D-loop) to TAP2 His661 (H-loop). The paper reports these loops moving apart about 3 Å after hydrolysis. Right of each PDB id: map resolution, Å.

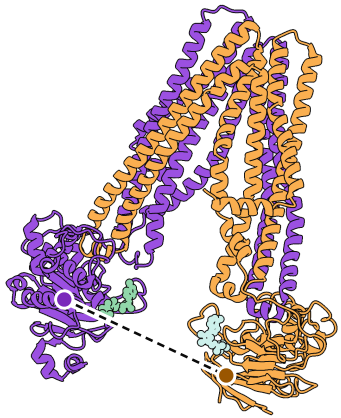
TAP1 TAP2 ATP ADP

peptide RRYQKSTEL (9 residues) Cα atoms of the gap

ER lumen

cytosol

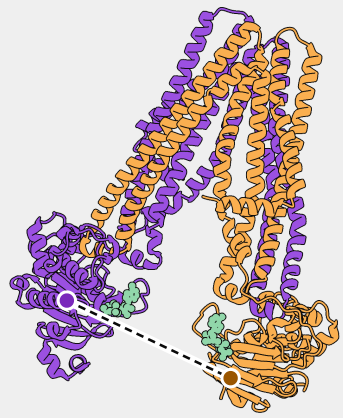
1 Inward-facing, ATP + ADP



9N65 3.7 Å
TAP1 ATP, TAP2 ADP
--- gap 55.1 Å

ADP out, ATP in

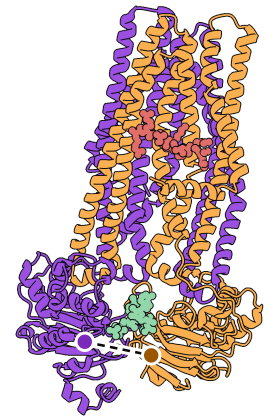
2 Inward-facing, EQ mutant



9N62 3.4 Å
TAP1 ATP, TAP2 ATP
--- gap 55.8 Å

peptide binds

3 Inward-facing, peptide bound

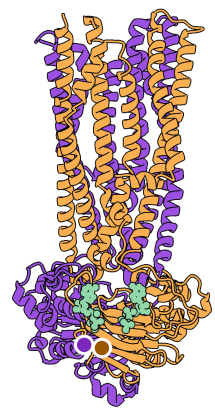


9N61 3.6 Å
TAP1 ATP, TAP2 ATP, peptide
--- gap 21.0 Å

NBDs dimerize, peptide released

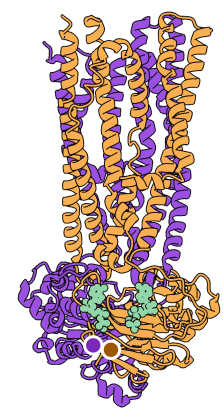
4 Outward-facing, two forms

4a Open, EQ mutant



9N63 3.1 Å
TAP1 ATP, TAP2 ATP
--- gap 5.8 Å

4b Kinked, EQ mutant



9N64 3.7 Å
TAP1 ATP, TAP2 ATP
--- gap 5.9 Å

ATP hydrolysed (wild type, next box)

How to read

Arrows join separate samples, not the steps of one molecule. Numbers give the reading order only.

Grey box: order not given in the paper; peptide and ATP bind independently.

EQ mutant (entry title): TAP2 catalytic Glu changed to Gln; ATP hydrolysis diminished.

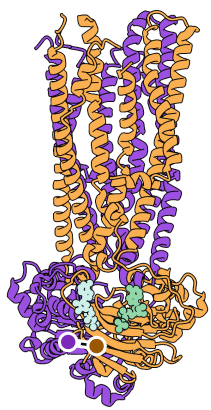
Open and kinked: the entries' names for two forms of TAP2 helix 3.

Inward-facing: cavity open to the cytosol. Outward-facing: open to the ER lumen.

Superposed on the TAP2 transmembrane core; dots and ligands are drawn in front.

NBDs separate

5 Outward-facing, ATP + ADP



9N66 3.2 Å
TAP1 ATP, TAP2 ADP
--- gap 9.4 Å

Separate cryo-EM samples. Order follows the paper's cycle, except box 2, which the paper does not place.

After Fig. 1A, 2A, 4A, 5A of Lee et al., Immunity 2025, doi:10.1016/j.immuni.2025.08.003, CC BY 4.0. Data: PDB 9N61-66. Redesigned; gaps computed, rest as in entries.